

Aegis: Privacy-Preserving On-Device Combat Motion Intelligence

Snapdragon AI Lab Build & Present Challenge Submission | Technical Description

Participant: Aniket Kapgate	Target Platform: Snapdragon X Elite / Plus HP PCs
Repository: github.com/aniketkapgate7-crypto/Aegis-Fig-ht-Pattern-Analyzer	Status: Verified MediaPipe CPU Baseline QNN Optimization Planned

1. Executive Summary

Aegis is an on-device, privacy-preserving motion intelligence application designed to deliver real-time, explainable combat and athletic movement analysis on Snapdragon-powered HP PCs. By combining normalized skeletal pose landmarks with a sliding-window temporal kinematic engine, Aegis detects rapid strikes, defensive guards, sudden approach, and fall patterns directly on the local workstation. It operates completely offline, eliminates cloud video streaming vulnerabilities, and empowers human operators with actionable, explainable telemetry. Aegis does not perform face recognition, face identification, facial embeddings or biometric identity tracking. One coarse nose keypoint is used only for pose geometry.

2. Problem Statement

- **Reactive Post-Incident Review:** Incidents and athletic form breakdowns are typically reviewed hours after occurrence from passive recordings, failing to prevent injury.
- **Cloud Latency & Network Fragility:** Streaming raw video to cloud servers introduces latency (200-800ms), consumes network bandwidth, and fails entirely during connectivity interruptions.
- **Biometric & Privacy Vulnerabilities:** Transmitting video over public networks exposes individuals to facial recognition, identity harvesting, and severe privacy violations.
- **Opaque Classification:** Conventional vision models output unexplainable classification scores without identifying which limb or motion triggered the alert.

3. Proposed Solution

Aegis resolves these issues through an edge-native, human-in-the-loop movement intelligence pipeline: **100% On-Device Processing** processes frames in local RAM without cloud transmission. **Skeletal Landmark Abstraction** converts pixels into 9 normalized joint coordinates. Aegis does not perform face recognition, face identification, facial embeddings or biometric identity tracking. One coarse nose keypoint is used only for pose geometry. **Explainable Kinematics** evaluates multi-frame limb reach and velocity. **AEGIS CORE HUD** displays live FPS, latency, threat meters, and optional deduplicated incident logs.

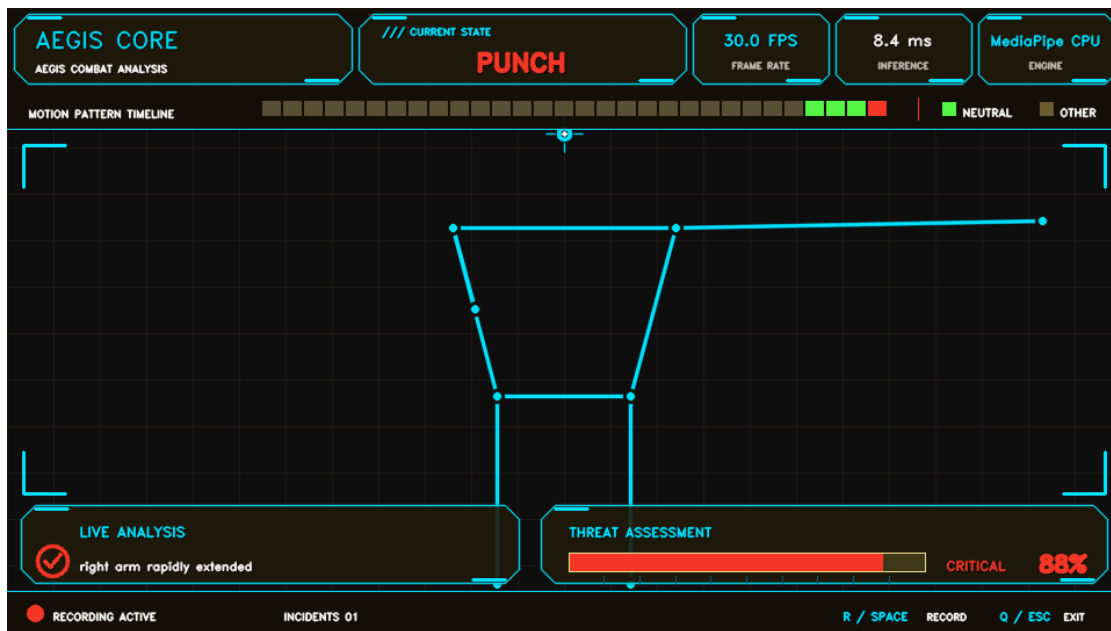


Figure 1: AEGIS CORE live heads-up display operating at 960x540 with synthetic martial arts pose telemetry.

4. Target Users & Use Cases

- **Combat Sports & Athletic Coaching:** Quantitative feedback on punch extension speed, guard discipline, and kick reach in boxing and martial arts training.
- **Supervised Facility Safety:** Assisting athletic supervisors in monitoring training mats to detect sudden falls, physical trauma, or aggressive collisions with human-in-the-loop oversight.
- **Edge AI & NPU Research:** Demonstrating how Snapdragon Hexagon NPUs can offload computer vision tasks from host CPU cores efficiently.

5. System Architecture & Technical Implementation

Aegis employs a strictly decoupled, modular architecture: **Frame Acquisition:** OpenCV captures video stream, resizing to 960x540. **Pose Estimator:** MediaPipe Pose generates 9 keypoints with visibility gating (>0.40). **Temporal Window:** A 12-frame deque tracks joint coordinates over time. **Pattern Engine:** Evaluates kinematic rules (Fall > Punch > Kick > Approach > Guard > Neutral). **Threat Fusion:** Combines signals into a 0.0-1.0 threat index. **AEGIS CORE UI:** Renders HUD and manages deduplicated JSONL incident logging.

6. Pattern-Detection Methodology

Pattern	Kinematic Criteria	Threat	Evidence String
Neutral	Equilibrium, no thresholds crossed	0.05	no risk-pattern threshold crossed
Guard	Wrists within 1.30x shoulder width of nose	0.30	both hands raised near face
Punch	Arm reach >1.45x, speed >0.90/s	0.88	right arm rapidly extended (reach 1.65x)
Kick	Leg reach >1.35x, speed >0.65/s, elevated	0.90	right leg extended, ankle elevated
Rapid Approach	Torso scale growth >6% across 3 frames	0.72	body scale increased rapidly (+12.4%)
Fall	Torso horizontal ratio >1.25, compact	0.90	torso horizontal ratio 1.84

7. Privacy, Safeguards & Responsible-Use

- **Zero Cloud Streaming:** Video is processed in volatile memory and immediately overwritten.
- **No Facial Biometrics:** Aegis does not perform face recognition, face identification, facial embeddings or biometric identity tracking. One coarse nose keypoint is used only for pose geometry.
- **Opt-In Local Logging:** Logging occurs only when enabled by the operator (R or Space key).
- **Human Interpretation Required:** Aegis outputs motion metrics, not intent or guilt. Autonomous enforcement is strictly prohibited.

8. Verified Evidence & Snapdragon Optimization Plan

Current Verified Baseline: Functional webcam ingestion, 17/17 deterministic unit tests passing in pytest, and truthful provider reporting (MediaPipe CPU is the only currently verified active pose provider). Supplying AEGIS_MODEL_PATH does not connect model outputs to pose estimation; runtime diagnostics only verify provider availability. **Planned Snapdragon Migration Pathway:** 1) Select compatible pose model (YOLOv8n-pose or RTMPose) from Qualcomm AI Hub. 2) Compile and quantize (INT8) targeting Hexagon NPU on Snapdragon X Elite/Plus HP laptops. 3) Genuine QNN pose execution requires an ONNX/QNN pose-estimator adapter that preprocesses frames, calls session.run(), converts outputs to normalized PoseFrame landmarks, and replaces MediaPipePoseEstimator via QNNExecutionProvider. 4) Run scripts/benchmark.py on an identical clip to capture comparative FPS and p95 latency gains.

9. Repository & Challenge Status

GitHub Repository: <https://github.com/aniketkapgate7-crypto/Aegis-Fight-Pattern-Analyzer>
Branch: feat/snapdragon-submission-readiness | **Challenge Track:** Build & Present